

Gender identity, household income composition, and the decision to work

Allegra Saggese

October 7, 2026

Women have made substantial gains in participation in the labor market in the US, driven primarily by married women entering the workforce during from the 1970s to 2000s. More recently, labor force participation growth has plateaued and womens' income and work hours have persisted below mens'. This research presents an argument attributing the recent stagnation in womens' work as a consequence of constraints imposed by broadly accepted social norms. Men and women take on a shared identity as a couple upon marriage, and there is a penalty imposed on working couples where a wife outearns her husband (the breadwinner norm). I derive early evidence the norm varies with household wealth, and is enacted on within households. The norm is difficult to detect as female labor force participation (extensive margin) has historically increased, but evidence is seen in a married woman's choice of work hours (intensive margin). Using IPUMS data from 1980 to 2024 linked to county political voting data I conduct a three part analysis. First I replicate the Bertrand, Kamenica, and Pan (2015) breadwinner norm analysis to demonstrate the intensity of the norm over time. Second, I document evidence to support the norm responds to household wealth. Third, I rationalize these facts with a household model in which an identity cost of violating the breadwinner norm generates a gap between predicted and worked hours of wives. The model suggest the norm distorts labor supply, but requires further evidence of the strength of the response to household wealth. Future work warrants improved data on household wages and wealth measures.

University of California, Santa Cruz. Email: absaggese@ucsc.edu. I thank Pascal Michallait for book recommendations, feedback and general lively discussions on this topic and throughout this independent research process. I also extend my gratitude to Jeremy West and George Bulman who taught, and my colleagues, who attended, the applied microeconomics sequence presentations throughout the 2025-26 academic year. Their feedback was invaluable.

1. Introduction

Research question. Do persistent gender norms materialize in U.S. women’s labor-market decisions today, and is the norm stronger where households can afford to enforce it? The female labor force participation rate (LFPR) rose from 33% in 1950 to about 58% in 2024 and female wages continued to rise, yet gains stalled after 2000 and there is no country in which female LFPR exceeds male LFPR (Ramey and Francis 2009; Gethin and Saez 2025). Institutional barriers have fallen, demonstrating the extensive margin is largely won. The open question remains on how gender norms govern the intensive margin of women’s work.

Answer to the question. Norms bind on the intensive margin, and they bind unevenly. The conservative–liberal differential in female labor supply is concentrated at the top of the income distribution. The breadwinner norm has not disappeared, and there is evidence that it is still enacted, although the strength of the norm is demonstrable decreasing over time.

Contribution. This paper makes three contributions. First, a *new stylized fact*: the partisan gap in female labor supply is income-elastic, invisible in pooled averages and concentrated at Q5. Second, a *replication and extension* of Bertrand, Kamenica, and Pan (2015). I reproduce the bunching of couples below the equal-earnings threshold in national data and extend it to partisan heterogeneity and to a kink in wife’s hours. Third, a two-period model, departing from the literature’s fixed norm penalty by embedding an identity cost in the style of Akerlof and Kranton (2000). I test whether the model can explain the stylized facts, specifically, whether the norm demonstrates income-elastic properties.

Roadmap. Section 3 reviews what is known. Section 4 describes the data used in both the empirical analysis and for the fit of the model. Section 5.1 replicates and extends prior analysis by Bertrand, Kamenica, and Pan (2015). Section 6 develops the model and provides calibration to available public data on US married households. Section 7 addresses the findings from the empirical and theoretical analysis. Section 8 addresses limitation and alternative explanations with a conclusion.

2. Motivation

Women in the United States, and across the world have never worked more hours or made more income from paid labor than men. This issue is particularly acute in the United States, where, despite years of progress¹, women have still never reached parity with men

¹See Goldin (1990); Goldin and Katz (2002) for a review

in labor markets. The most clear evidence is held in the legal and regulatory protections that women were afforded from hiring to treatment in the workplace. But in the context of the US, legal protection of women in the home has lagged. Child marriage is banned in only thirteen of fifty US states, where three states have no minimum age requirement for marriage. Marital rape, while broadly punishable, varies in terms of severity by US state. And a cultural boomerang from the lean-in era of Sheryl Sandberg to the social media conquest of tradwives influencers raises concerns that while workplace opportunities are available to both women and men of merit, existing barriers may still be holding women's aggregate potential in the labor market back.

I argue that identity still plays an enormous role in the way women and men decide to work. And particular that the norms comprising a gender identity are important for both men and women. I argue a political *and cultural* focus on the role of women in society has obscured the importance of male identity, and the interconnectedness of male and female gender norms between married individuals. As individuals, we may have our own identities; but jointly in a couple, an additional expectation derived from gender roles is introduced.

The US is a useful setting for this question because its gender and family norms are unusually resistant by peer-country standards. Religiosity remains far above that of Western Europe; abortion access lost federal constitutional protection in *Dobbs v. Jackson* (2022); and American women begin childbearing markedly *earlier* than their European counterparts, with a mean age at first birth of roughly 27–28 years against approximately 31 in Italy and Spain and 30 in Germany (National Center for Health Statistics 2025). While the age of first childbirth is driven by an extraordinary tail in teen pregnancies not present in European economies, earlier first births in the higher-income country run against the standard opportunity cost logic where we would expect to see wealthier economies postponing childbearing furthest.

Gender roles, therefore, may actually take on a more tangible effect once two people are coupled up. I will test a simplified version of this hypothesis. I build on the existing literature around gender roles, looking particularly at married couples in the US and the norm's influence on women's decision to work. Secondly I ask whether the intensity of the penalty for violating the well-defined gender roles within a couple exhibit time variance and heterogeneity. The former part of the question will be evaluated by replicating the canonical analysis by Bertrand, Kamenica, and Pan (2015), using a ten-year update to the survey data. From this we will estimate the magnitude of the norm and evaluate the previous period of analysis against the most recent data to identify whether women reduce their work hours in response to their husband's work hours. This is known as the breadwinner's norm.

The core contribution of this research is to argue that while women's contribution to

wage labor have gained enormously along the *extensive margin*, their gains on the *intensive margin* remain stagnate in recent years. This is due to two core facts that I will show, including an intra-household utility function governed by the breadwinner norm, causing women to adjust their earned income downward. Secondly, I will demonstrate that the gains in female labor force participation are separate and unrelated to the mechanisms which govern the gender pay gap. The bright spot is that today women are much more likely to be at work when married relative to the past. Yet as more women enter a fair labor force, they may be facing a barrier from within. My findings suggest gender norms applied to married couples are consequential for aggregate labor market outcomes.

3. Literature review

The long run rise in U.S. female labor force participation (LFP) is one of facts in labor economics (Goldin 1990; Ramey and Francis 2009; Olivetti 2013). But the story has become less urgent. Female LFP has been broadly flat for the last two decades. The composition of women's employment, when women work and how intensely they work remain open areas of investigation (Goldin and Mitchell 2016; Goldin, Kerr, and Olivetti 2022). In parallel, labor force participation has been declining, albeit slowly, in the US, driven by a drop in male workers (U.S. Bureau of Labor Statistics 2026). In a global analysis of working hours, Gethin and Saez (2025) identify the long run substitutability between women and men's work on the intensive margin. In the case of the US, female workers' entry seems tethered male participation rates. Alternative explanations may focus on broader conditions. Regardless, the data in the US demonstrate a trend such that female LFP seems tethered to males' (see Figure 1). Female LFP has yet to crossover, or exceed LFP of men in any country in the world (International Labour Organization and World Bank 2026). Co-movement in the two seem to be dominate the pattern of the trend, as the recent extensive margin of female work has plateaued, after the rise in working women from the 1950s to 1990s was driven by new married women entering the market.

Historical explanations of womens' growing participation in the labor market span from healthcare to home appliance adoption. Household appliances raised the productivity of home production and released women's time (Greenwood, Seshadri, and Yorukoglu 2005) while contraception allowed the timing of births to be separated from the timing of careers (Goldin and Katz 2002). Both operate on constraints by lowering the cost of market work. Both predict monotone convergence once the constraint is relaxed. But, neither explain why convergence on the intensive margin has stalled, nor why it may vary between households where women have similar predicted wages.

The margin that has moved is relative, not absolute. Women's higher education attainment now exceeds men's, and in twenty two US metropolitan areas women under thirty out-earn their male counterparts (Pew Research Center 2022). Yet real wage growth has

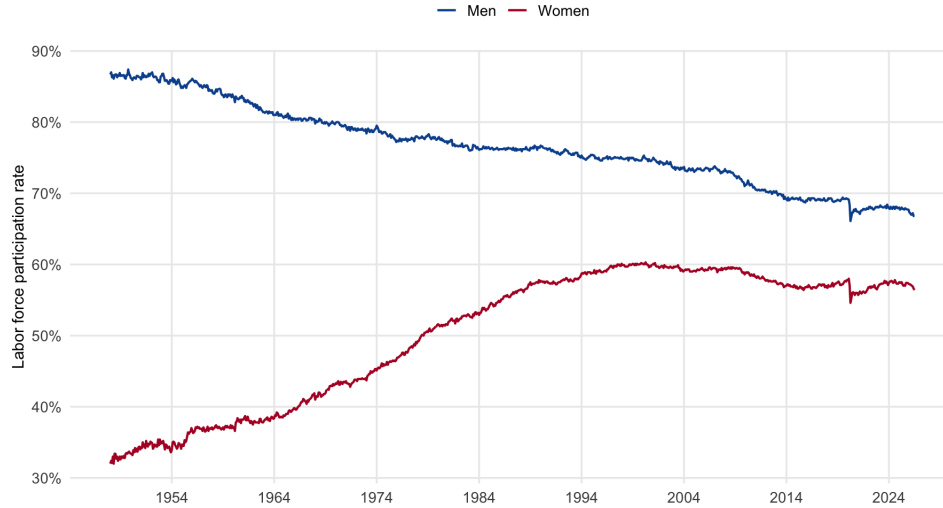


FIGURE 1. US LFPR by gender from 1950-2025 (BLS 2026)

been slow since the 1980s, and the gender pay gap has narrowed as much due to cohort turnover and men's positional losses as well as through women's gains (Arellano-Bover et al. 2025; Attanasio, Hurst, and Pistaferri 2013). While wages are stagnating, productivity continues to rise (Stansbury and Summers 2020). Putting productivity aside, this aggregate gender gap conceals a rising incidence of couples in which the wife's earnings potential meets or exceeds her husband's. I contribute to the macroeconomic literature on to intra-family household decision making, where a relative-income becomes consequential for households choose to allocate their time in home and market economy.

Female labor supply has demonstrated responsiveness to a number of historic events. Women have primarily shouldered the burden of the household adjustment, serving as a more elastic component of total household labor. In response to wartime mobilization, women picked up slack in the labor market at scale (Acemoglu, Autor, and Lyle 2004) and subsequently existed the market in the post-war period (Rose 2018). When a husband loses his job, wives often increase their labor supply as a response (Lundberg 1985; Blundell, Pistaferri, and Saporta-Eksten 2016; Melvin Stephens 2002).

Shocks to household wealth, as distinct from earnings, produce smaller but clearly gendered responses. There is a negative gradient between married women's working hours and housing asset value (Disney and Gathergood 2018; Rock, Kovaleva, and Potoms 2025). Increases in household wealth, in other words, are absorbed disproportionately by withdrawing the wife from the market, either because women have a lower reservation wage or, potentially the social expectation. Whether certain characteristics of households - location, number of children, or belief - influence this relationship is not well understood.

The consistent pattern across this literature is that the elasticity of female labor supply depends on which margin is shocked, and that no single elasticity summarizes it. Re-

sponses to aggregate demand and to a husband's job loss are significant while responses to changes in wealth are modest but gender-asymmetric. The response of relative wages between wives and husbands remains up for debate. Gender norms, if well understood, should help predict how women respond to their shifting position within the economy and within the household. This research will interrogate how social norms effect women's adjustment on the intensive margin using a theoretical application and with consideration to heterogeneity in the norm. Existing work to build on how wives adjust their margin of labor include Ko, Oh, and Zhang (2024); Lee (2025).

A second strand of literature is that of what motivates or governs economic choice of individuals. A core alternative that is underexplored is the belief that these documented preferences for work carry a social component. Akerlof and Kranton (2000) were the first to propose embedding identity directly in the utility function. The core idea is that an individual is assigned to a social category and incurs a loss from acting against that category's behavioral prescriptions. This concept changes the standard conclusions on the household division of labor because the imposed cost derived from the social norm is attached to who, based on their social category, performs the task, and not to who performs it *more cheaply*. Social norms are well documented in their impact on behavior, particularly with respect to gender (Kleven and Landaís 2017; Bursztyn, González, and Yanagizawa-Drott 2020).

This mechanism has also recently been applied data. Sakamoto and Kohara (2025) show that Japanese households holding conventional gender roles, as measured through household survey responses, allocate more female time to housework and less to market work with welfare consequences. Variation in belief intensity results in tangible differences between female labor within households. Lee (2025) introduce a social norm penalty on households who have a wife that works, where the implied tax on female earnings is derived from a matched unmarried woman in the sample.

The breadwinner norm, or the belief that husbands should outearn their wives is the subject of Bertrand, Kamenica, and Pan (2015), who document a sharp drop in the density of couples just past the point where the wife's share of household earnings reaches one half. The analysis concludes an aversion to violating a well established gender norm within couples. The authors report corroborating effects on marriage formation, female participation, wife's earnings conditional on working, and the division of housework. Since its publication ten years on, there have been substantial counterclaims made to disentangle the bunching demonstrated. Zinovyeva and Tverdostup (2021) argue the discontinuity is mechanical rather than normative using Finnish data linking employer to employee to show that bunching is a result of earnings convergence among couples working at the same firm. Binder and Lam (2018) introduce a secondary market, that for marriage, to show that selection alone generates the observed equilibrium patterns.

Inferring preferences for labor does not hold where matching and sorting into more equal income couples is occurring in parallel. Finally, Rosenberg (2024) conducts a near replication of the study. The replication includes a conventional adjustment where hourly wages is assessed in place of annual earnings, and predicted earnings for a husband are also used. The results demonstrate a sign reversal a precise null under couple fixed effects.

Taken together these papers leave the norm interpretation contested rather than refuted. The spirited confrontation to the original analysis by Bertrand, Kamenica, and Pan (2015) is a testament to the lack of consensus on the issue. Despite progress, why do women still persistently work less than men? And how should begin to measure women's successes and existing constraints in the labor market? This analysis will interrogate whether the relationship between relative wages within a household, as measured by income is best described as a fixed belief in identity or by a norm that fluctuates with circumstance. Section 5.1 reports the most current replication of Bertrand, Kamenica, and Pan (2015), and the model in Section 6 rationalizes what survives.

My research is formed using a collective household model in the style of Chiappori (1992), with the absence of intra-household bargaining, and builds upon the work of (Ko, Oh, and Zhang 2024) where identity is incorporated into a household model such that a breadwinner penalty is introduced when wives' earnings share exceeds a threshold δ . While a persistent norm and rising intra-household bargaining power of wives reproduces the pattern of wage-driven growth in participation, followed by stagnation, the use of a threshold (binary) penalty differs in flexibility. The core contribution to this body of existing research is that the relative wage gradient between couples, rather than an indicator on a shared threshold, is an essential mechanism for understanding female labor adjustment on the intensive margin. This research will work to present a one-period static model that complements rather than rivals both Ko, Oh, and Zhang (2024) and more recently, Wu (2020).

4. Data

Data are drawn from IPUMS USA, accessed through the IPUMS API. The extract comprises 109,111,431 person records across twenty-nine samples: the decennial censuses of 1970, 1980, 1990 and 2000, and American Community Survey (ACS) single-year files for every year from 2001 through 2024. The ACS is a repeated cross-section; single-year files are published for geographies of at least 65,000 residents, with three- and five-year files covering smaller areas at the cost of currency. Because the analysis requires the most recent years, and because no household is observed twice in any case, single-year files are used throughout. Details of the extract are in Appendix A.

Cultural context is proxied by county presidential vote margins from the MIT Election Data and Science Lab, hosted on the Harvard Dataverse. Presidential returns are available

for the elections of 2012, 2016 and 2020, and are carried forward to intervening years, so the political data bound that portion of the analysis to 2012–2020.

Key variables. Couples are opposite-sex spouse pairs in which both partners are of working age, 18 to 65. Table 1 lists the variables extracted and their use.

TABLE 1. ACS variables

Group	Variables	Used for
Identifiers	YEAR, SAMPLE, SERIAL, PER- NUM	Person and household keys
Weights	HHWT, PERWT	All weighted statistics
Linkage	SPLOC	Spouse pointer to identify couples
Demographics	AGE, SEX, RACE, HISPAN, EDUC, MARST	Sample restrictions, demographic cells
Labor	EMPSTAT, LABFORCE, UHR- SWORK, OCC, IND	Labor market participation, hours
Weeks	WKSWORK1, WKSWORK2	Annual hours
Income	INCWAGE, INCBUS, INCFARM, IN- CBUS00	Labor income
Income (other)	INCTOT, INCINVST, INCOTHER, INCSS, INCWELFR, INCRETIR	Non-labor income; robustness
Geography	STATEICP, COUNTYICP, METRO	Fixed effects; county merges
Household	NCHILD, YNGCH, ELDCH, GQ, MULTGEN, HHINCOME	Controls

Sample construction. The three components of the paper use different samples (Section 5.1, 5.2, and 6). I will explicitly state the relationship between all three samples. All the data is derived from the ACS survey data, yet samples are sliced or certain variables are selected based on measurement choice. The broadest sample of data is presented in Table 2.

To start, couples are formed by matching on YEAR, SAMPLE and SERIAL and retaining only mutually linked spouses, so that each partner’s SPLOC points at the other. This is conducted for all sub-samples. This yields 19,232,697 couples across 1970–2024, of which 16,415,647 have a husband in work. Because the data are repeated cross-sections, no couple appears twice and nothing in the analysis is a within-couple change. One caveat follows from the linking rule: SPLOC identifies IPUMS “probable spouses,” which include cohabiting partners, and marital status is not imposed as a further restriction. Roughly 88.5% of the resulting couples are legally married with spouse present; the remainder are predominantly cohabiting. We include cohabiting couples in line with the most up to

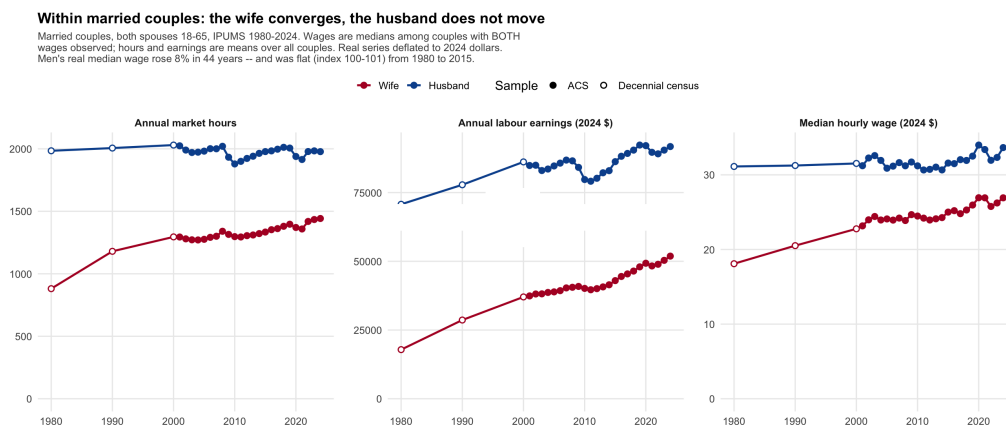


FIGURE 2. Wife and husband nominal labor income (1970-2024)

date literature (Lee 2025).

The replication analysis of Section 5.1 follows Bertrand, Kamenica, and Pan (2015) in restricting to couples in which the husband works, and estimates on a random subsample of 250,000 observations per specification to keep the demographic-cell construction tractable; the reported standard errors account for this. The *model* of Section 6 uses 16,894,715 couples over 1980–2024. The 1970 census is excluded because INCINVST is not collected that year, so non-labor income cannot be constructed.

The measurement of the norm elasticity in Section 5.2 is the most restricted, at 2,755,378 couples, and falls to 2,477,474 in the regressions, which additionally require positive husband’s labor income to form the income-decile fixed effects. There are two restrictions which bind. The political data limit the window to 2012–2020. More consequentially, IPUMS identifies county of residence only for sufficiently populous areas, so the sample spans 430 counties in 45 states. Coverage is therefore skewed urban and suburban: only 32.5% of the sample couples live in Republican majority counties, though such counties are roughly four-fifths of all counties nationally. The sample should not be taken as representative of the US population, but produces results which can be estimated for comparison between similar size counties. As suggested, estimates using the political proxy should be read as applying to large counties, and the rural areas where traditional norms may be strongest are largely absent by construction.

Measurement. Labor income combines wages and self-employment. The choice of selecting labor income is non-trivial, such that surveys have a number of methods for asking what weekly earnings were, weekly hours are, and annual wage income. We start with the most clear variables in the data including wages and self-employment wages. Self-employment income is INCBUS00 where it is collected, from 2000 onward, and INCBUS + INCFARM in the earlier censuses, where farm income is reported separately. labor income is then

$$y = \max(\text{INCWAGE} + \text{selfemp}, 0),$$

so that reported business or farm losses cannot produce negative labor income. IPUMS codes missing and inapplicable values as large sentinels (INCWAGE = 999999, and similarly for the other income variables). These are set to missing before any summation.²

Three exclusions apply where an implied hourly wage would not be meaningful. First, a wage is computed only for individuals working at least 260 hours a year, roughly five hours a week: below that threshold the implied wage is dominated by reporting error, with wives working one to fourteen hours a week showing a mean implied wage of \$81 against \$29 at full time. This implies self-reported wage data is not properly capturing wage work.

²Note most observations with INCWAGE = 999999 are for under-16s and are removed from the sample with the age filter.

This type of work, in a manual review of occupation codes, yielded a number of board or executive positions. Second, couples in which either spouse reports at least \$100,000 of labor income on fewer than 260 hours are dropped entirely; these are payouts and returns to capital rather than wages, and they constitute about one-tenth of one percent of observations. This is due to an extreme value case that does not well capture the decision to work additional hours. Third, implied wages below half the prevailing federal minimum wage for that year are treated as unmeasured, and the top percentile is trimmed within year. Both bounds are year-specific rather than fixed in dollars, since a common threshold is meaningless across a period in which the federal minimum rose from \$1.60 to \$7.25.

Annual hours are then calculated as the product of usual weekly hours and weeks worked. WKSWORK1 reports weeks directly, but is unavailable between 2008 and 2018, when the ACS collected weeks only in intervals. For those years weeks are reconstructed from WKSWORK2 using the conditional mean of each interval (7.35, 20.97, 32.98, 42.33, 48.21, 51.86) rather than the interval midpoint. Reconstruction affects twenty four of the pooled sample, and an indicator is retained so that results can be checked against the direct measure in the years where both exist.

Summary statistics. Table 2 reports means for the period covered by Bertrand, Kamenica, and Pan (2015) and for the years since. The changes that can be seen are that wives are older and better educated than their husbands by the later period, their participation is nine points higher, and their share of couple labor income rises from 0.275 to 0.306. The share of couples in which the wife out-earns her husband rises from 17.3 to 21.9. Husbands' weekly hours are flat throughout, a point returned to in Section 6.

TABLE 2. Average summary statistics by years

Panel A: Demographics					
Era	Wife Age	Husband Age	College (W) (%)	College (H) (%)	Wife LFP (%)
BKP era (1970–2011)	40.6	42.7	28.8	31.0	69.8
Post-BKP (2012–2024)	42.4	44.3	41.8	38.2	74.4

Panel B: Labor Supply and Income					
Era	Hours (W)	Hours (H)	L income (W)	L income (H)	Wife Share (%)
BKP era (1970–2011)	27.6	44.4	21,310	51,750	27.5
Post-BKP (2012–2024)	28.5	43.8	38,018	80,523	30.6

5. Empirical approach

5.1. Replication and extension of Bertrand, Kamenica, and Pan (2015)

Bertrand, Kamenica, and Pan (2015) document that the distribution of the wife's share of couple income falls sharply just above one half. The breadwinner norm is a consequence of couples avoiding a marriage in which the wife out-earns her husband. The identification of the consequences of that norm rests on a constructed regressor, and because both the regressor and the treatment of the threshold matter for the analysis. I first review the selection of data and the design of the empirics before moving to results.

The potential-earnings regressor. The realized income share is endogenous to the labor supply decision it is meant to explain. Bertrand, Kamenica, and Pan (2015) therefore replace it with the probability that the wife would out-earn her husband were her earnings drawn at random from women like her. Each wife is assigned a demographic cell defined by five-year age band, education group, race and state of residence in a given year. Within each cell I take the earnings distribution of working women and evaluate it at nineteen percentiles, the fifth through the ninety-fifth.³ Writing y_j^p for the p th percentile of wife j 's cell and y_j^m for her husband's realized income,

$$(1) \quad \text{Pr}(\text{wife earns more})_j = \frac{1}{19} \sum_p \mathbf{1}[y_j^p > y_j^m]$$

The regressor varies with the husband's income and with the wife's earnings capacity, but not with her realized labor supply. The dependent variable in the second specification is the income gap, $(y_j - \bar{y}_j) / \bar{y}_j$, where $\bar{y}_j$ is the mean of her potential distribution; it is defined only for working wives, which accounts for the smaller sample in that column.

Treatment of the threshold. Reported income shares mass at exactly one half. This occurs due to rounding, survey bias or survey imputation, rather than from behavior. Following Bertrand, Kamenica, and Pan (2015, fn. 12) I redistribute that mass across neighboring bins using a triangular kernel, so that bin k of n receives weight $(n/2 - |n/2 - (k - 1)|) / (\frac{n}{2}(\frac{n}{2} - 1))$. For the discontinuity estimates I additionally exclude a two-percentage-point donut around one half, so that the comparison is between densities on either side rather than at the point where measurement error concentrates.

Specification. Each specification regresses the outcome on the constructed probability (1), controlling for the husband's income, vigintiles (20-quantile) of the wife's potential income,

³Exclusion of outlier percentiles is in line with the previous author's choice.

five-year age bands and education groups for both spouses, race, and state and year fixed effects. Estimates are weighted and standard errors are clustered on the demographic cell. The preferred specification enters the husband's income as a cubic as linear term constrains the relationship in a way the data reject. Note this specification choice is a source of the apparent discrepancy between the first and second rows of Table 3.

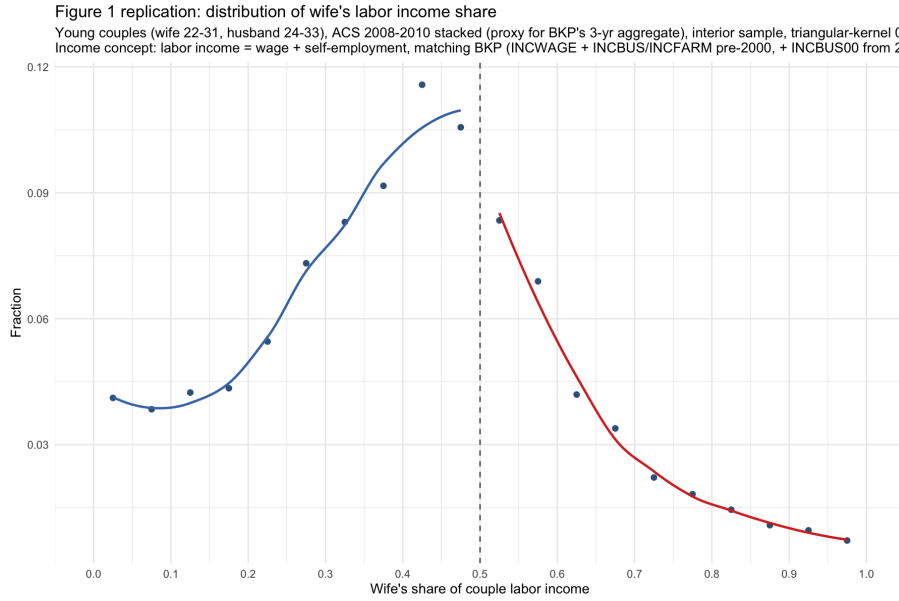


FIGURE 3. Distribution of the wife's share of couple earnings

Notes: Density of the wife's share of couple labor earnings on the interior sample, twenty bins, with the exact-one-half mass redistributed by triangular kernel. The discontinuity at one half is the fact the analysis begins from.

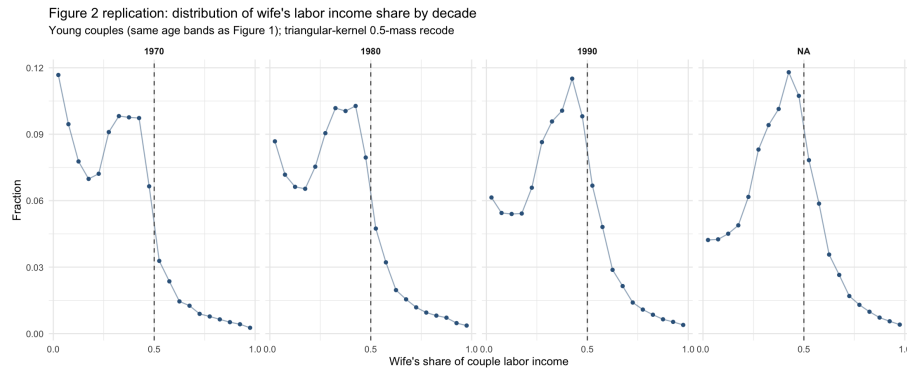


FIGURE 4. Distribution of the wife's share of couple earnings over time

Replication. Table 3 reports the estimates for the period covered by Bertrand, Kamenica, and Pan (2015). The constructed probability enters negatively and significantly in both specifications, corroborating the findings in the previous analysis. A wife that is more

likely to out-earn her husband is less likely to work, and conditional on working earns less relative to her potential. The overall replication produces well, as the data and methodology selected were as close to the original paper as possible. The share of couples in which the wife earns more is 25.6% against the 26% reported by Bertrand, Kamenica, and Pan (2015); married women's participation runs 43.0%, 70.6% and 74.0% in 1970, 1990 and 2010, against their approximate 44%, 70% and 74%; and the density in Figure 3 reproduces the discontinuity at one half. The estimates are somewhat smaller in magnitude than the published coefficients as expected. The core difference between the replication and the original analysis was the choice to use single-year samples throughout rather than the pooled three-year files available to the original authors for 2008 to 2013.

TABLE 3. Replication: BKP Era (1970–2010)

	Wife LFP	Income Gap
ln(Husband Income)	0.0577*** (0.0071)	-0.3021*** (0.0175)
Cubic ln(Husband Income)	-0.1329*** (0.0110)	-0.1590*** (0.0293)
Cubic ln(Husband Income) + Any Children	-0.1424*** (0.0109)	-0.1781*** (0.0292)
Observations	249,993	153,695
Full sample	8,079,239	8,079,239

Standard errors in parentheses.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Preferred specification: cubic ln(Husband Income).

Extension. Table 4 applies the identical specification to 2012–2024, years unavailable when Bertrand, Kamenica, and Pan (2015) was written. The comparison is the point of the exercise, and the two margins move very differently. In the preferred specification the participation response falls from -0.1424 to -0.0894 , a decline of 37%. The income-gap response falls from -0.1781 to -0.1727 , a decline of 3%, which is within sampling error of no change at all.

The takeaway from the replication analysis is the discovery that the norm is elastic on one margin and not the other. The norm documented by Bertrand, Kamenica, and Pan (2015) has not just weakened (or evolved) over time. Its effect on whether a wife works has fallen by more than a third in the years since their sample ended, while its effect on how much she earns relative to her capacity, conditional on working, is unchanged. Therefore, the declining gender norm in only a single parameter would not be able to generate

TABLE 4. Post-BKP Period: 2012–2024

	Wife LFP	Income Gap
ln(Husband Income)	-0.0212*** (0.0073)	-0.2603*** (0.0185)
Cubic ln(Husband Income)	-0.0828*** (0.0085)	-0.1652*** (0.0229)
Cubic ln(Husband Income) + Any Children	-0.0894*** (0.0085)	-0.1727*** (0.0229)
Observations	250,000	180,557
Full sample	4,979,852	4,979,852

Standard errors in parentheses.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Preferred specification: cubic ln(Husband Income).

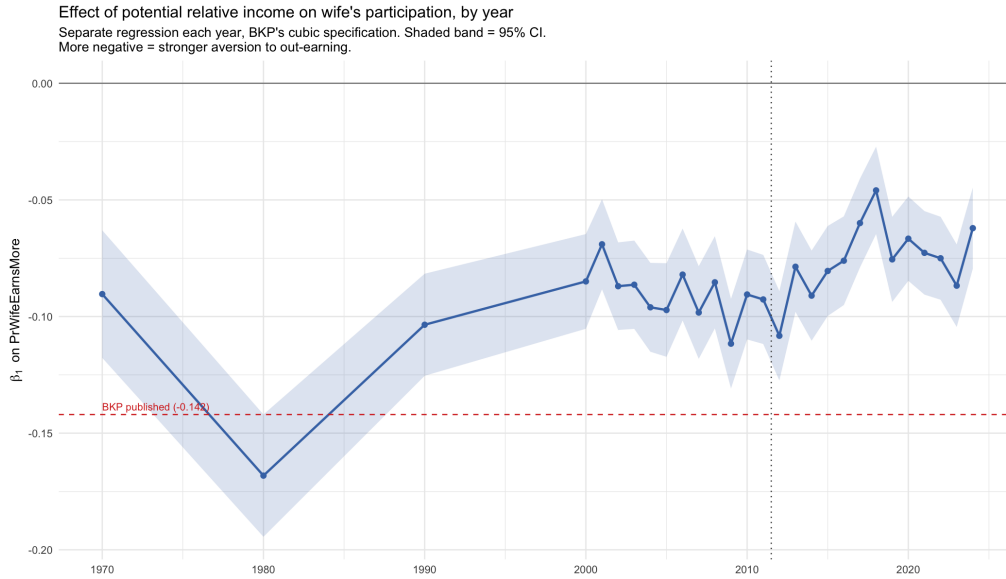


FIGURE 5. The two margins of the norm over time

Notes: Coefficient on the constructed probability (1), estimated separately by year in the preferred specification, with 95% confidence intervals. The participation response attenuates over the sample; the income-gap response does not.

that pattern. A uniform weakening would attenuate both margins at the same time. This asymmetry motivates the rest of the analysis. Findings suggests two prescriptions rather than just one: an expectation about married women’s participation, which has eroded substantially, and an expectation about relative standing within the couple, which has not. The empirical analysis in Section 5.2 asks which households does the relative-income norm bind on. The model in Section 6 is a structural approach to generating the observed bunching below the equal share income, and how that intensity may evolve over household characteristics. The relevant elasticity throughout is with respect to the couple’s resources rather than to time: a norm about relative position is a constraint households can only act on once they can afford to forgo the wife’s earnings. In conclusion we have confirmed the norm exists and remains persistent on its influence on relative-income norm.

5.2. Estimating the norm’s response to wealth

Is the norm that governs women’s choice of labor hours wealth-elastic? The literature demonstrates that labor supply is elastic to relative household wealth (Cesarini et al. 2017; Rock, Kovaleva, and Potoms 2025), wealth allows for consumption smoothing over time (Wickens 2012), women adjust in response to shocks. Wealth, similarly, is not a short term income shock, and elicits a different response than an adjustment in income. We consider wealth, as well as traditional cultural beliefs, as two explanatory forces for driving heterogeneity in the size or realization of the norm:

- *Culture*: areas that are have strong social beliefs identify with conservative values, one of which is traditional gender ideology
- *Income-elasticity*: household preferences are realizable once an individual can afford to forgo earnings for consumption

In order to test the elasticity of the norm to these two parameters, I generate a proxy for both wealth and social conservatism to test the responsiveness of the norm. To create a simple proxy for the strength of the norm intensity, I used the political election data at the county level and identified counties where the vote margin was favorable towards Republicans, the conservative party in the US, in recent elections. Counties were then labeled as conservative and progressive, respectively.⁴ The variable is named *conserv* in the regression in Table 5.

⁴While political preference is an imperfect measure of beliefs in traditional roles, it is a simple proxy that allows for county-level analysis of households. Similarly, more recent election in the United States have focused heavily on cultural issues, including preserving the nuclear family unit, banning abortion and anti-LGBTQIA+ advocacy. While these may not be explicit indicators of the strength of gender roles, we make the assumption that strongly conservative voters tend to favor or believe in structured gender norms. Consequently, we assume more progressive counties would impose a lower penalty on couples that do not adhere to the breadwinner norm, either due to less social pressure or increased acceptance of non-traditional beliefs.

To generate a strong proxy for the classification of a household as wealthy, simple data was not appropriate for use. Given housing carries both a debt and equity component which are oppositional on the total effect of wealth, I use a within-county decile ranking for households. Of the spousal household records linked to county identifiers and the vote margin data, I create a binary indicator (0, 1) for wealthy households such that those identified as owning a housing asset in the top quartile of reported house values for their state are classified as wealthy. This indicator also restricts on the house being owner-occupied. Finally, an indicator variable of whether the house is owned outright, outright is included.

Voting data were then merged using county identifying codes, FIPS, for all household records in the IPUMS ACS with FIPS. In total 430 counties across 45 states are included. Data were merged using FIPS and YEAR as the key. The empirical approach serves as a bridge between the replication study and the theoretical model. Two left hand side variables are identified: labor force participation by wives (extensive margin) and average weekly hours worked (intensive margin). Both specifications contain the same regressors, fixed effects and controls. The estimate is at the household, l , level.

$$(2) \quad FLFP_l = \beta_1 \text{Cons}_l + \beta_2 \text{Wealthy}_l + \beta_3 (\text{Cons}_l \times \text{Wealth}_l) + \mathbf{X}'_l \boldsymbol{\gamma} + \alpha_s + \delta_{\text{yr}} + \lambda_{\text{decile}} + \varepsilon_l$$

$$(3) \quad \text{HRS}_l = \beta_1 \text{Cons}_l + \beta_2 \text{Wealthy}_l + \beta_3 (\text{Cons}_l \times \text{Wealth}_l) + \mathbf{X}'_l \boldsymbol{\gamma} + \alpha_s + \delta_{\text{yr}} + \lambda_{\text{decile}} + \varepsilon_l$$

Controls included terms to control for age and college attendance for both male and female partners, in addition to the number of children and the log of labor income at the household level.

$$(4) \quad \mathbf{X}_l = f_{\text{emage}} + f_{\text{emage}}^2 + m_{\text{alage}} + m_{\text{alage}}^2 + f_{\text{college}} + m_{\text{college}} + n_{\text{child}} + n_{\text{child}}^2 + \ln(\text{labinc})$$

Table 5 reports the association between county political composition, household wealth, and married women's labor supply for the 2,477,474 couples observed in the ACS between 2012 and 2020. Data are retained where the male spouse has reported income that is non-zero. $\hat{\beta}_i$ are reported conditional on both spouses' ages and education, the number of children, the husband's labor-income decile, and state and year fixed effects. Standard errors are clustered by county. The results indicate that neither political context nor wealth alone is associated with lower female participation or working hours.⁵

⁵Additional analysis was conducted, and results are robust to education of wives, demonstrating wealth may have the dominating effect such that constraints are relaxed and norms can be realized (see Table A9 for robustness tests and alternative specifications.)

TABLE 5. Regression for norm elasticity estimate

Outcome	LFP (0/1)		Weekly wife hours		LFP (0/1)	
Model:	(1)	(2)	(3)	(4)	(5)	(6)
<i>Variables</i>						
conserv	0.0056 (0.0036)	0.0088** (0.0036)	0.4681*** (0.1403)	0.0071** (0.0034)	0.0078** (0.0037)	-0.0041 (0.0029)
wealthy	-0.0229*** (0.0038)	0.0275*** (0.0036)	1.700*** (0.1644)			-0.0640*** (0.0030)
conserv × wealthy	-0.0224*** (0.0055)	-0.0279*** (0.0056)	-1.486*** (0.2583)			-0.0201*** (0.0051)
outright				-0.0571*** (0.0024)		
conserv × outright				-0.0164*** (0.0035)		
wealthy_asset					-0.0054*** (0.0020)	
conserv × wealthy_asset					-0.0226*** (0.0041)	
<i>Fixed-effects</i>						
state	Yes	Yes	Yes	Yes	Yes	Yes
year	Yes	Yes	Yes	Yes	Yes	Yes
decile		Yes	Yes	Yes	Yes	
diagnostic decile						Yes
<i>Fit statistics</i>						
Observations	2,755,378	2,477,474	2,477,474	2,477,474	2,477,474	2,651,607
R ²	0.06534	0.07340	0.09355	0.07524	0.07327	0.18477
Within R ²	0.05820	0.05592	0.07587	0.05779	0.05578	0.12137

Clustered (fips) standard-errors in parentheses

Signif. Codes: ***: 0.01, **: 0.05, *: 0.1

5.3. Effect on extensive margin

The analysis residence in a Republican majority county predicts a 0.88 percentage point higher participation rate at a 5% significance level. Quartile home predicts a 2.75 point higher rate. The negative association appears only in their interaction, at 2.79 points, and reoccurs with the same sign and significance across three independently constructed wealth measures topquartile house value (2.79), outright (1.64), and reported asset income (2.26). Estimated separately within each of the nine sample years, the interaction ranges from negative 1.97 to 3.63 points and is distinguishable from zero in every one. The two housing measures diverge to provide us with key information on the positive coefficient on high house value. One reflects leveraged wealth, or debt, rather than wealth. Since a costly mortgaged home may require a second income to service, outright ownership infers the equity has been restored in the asset. The difference between the two is substantial. Outright ownership predicts 5.71 points lower female participation in the labor force. Taken together these estimates are inconsistent with a norm operating as a fixed cultural attribute, since conservative context alone predicts no withdrawal, and is equally inconsistent with a pure income effect, since wealth alone does predicts more work rather than less. They are consistent instead with a latent, income-elastic norm: a traditional preference over the intrahousehold division of market work that households express only once they hold the resources to act on it.

5.4. Intensive margin

To understand the relative adjustment women may make when choosing to work more, we evaluate the effect of living in a conservative area and being a part of a wealthy household on weekly hours. Results suggest a difference in female hours worked between Republican-majority and non-Republican-majority counties of about 1.5 hours less for wealthy households than for non-wealthy households, conditional on control variables. Parsing out the difference, wealthy households report 1.02 hours less worked by a wife while conserv is associated with significant increase in .47 hours. To interpret the channels of effect, even with state fixed effects, we may be cautious of estimating the β parameter on conservative given the presence of omitted variable bias. While state fixed effects control for regulatory or industrial policies at the state level, there is significant geographic differences in the labor market that are not controlled for within this sample. Republican counties tend to be more rural, for example, and we believe the results may skew as a result. Given the bias would be upwards against the norm, we can confidently report the interaction term as a lower bound.

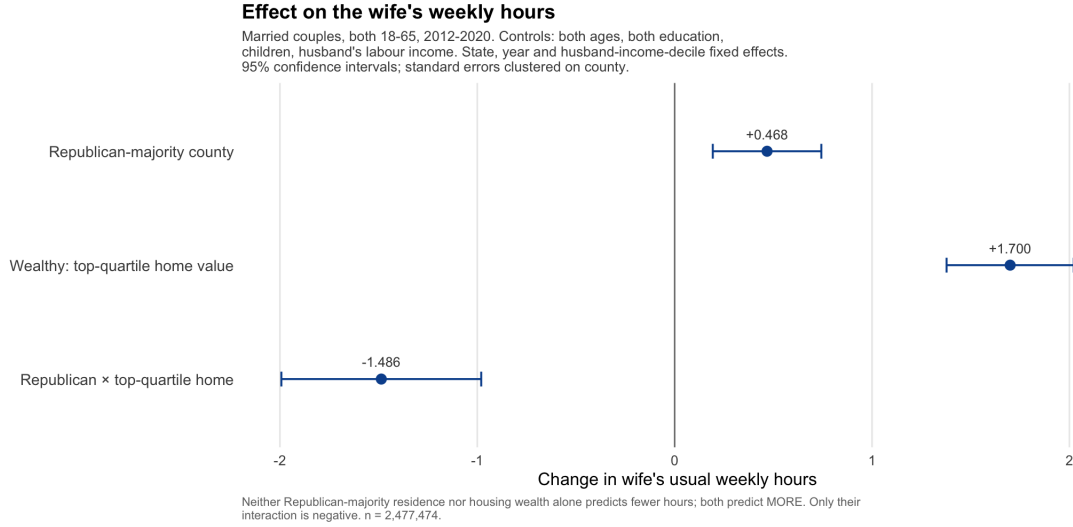


FIGURE 6. Coefficient plot of norm-indicator on wife's weekly hours

6. Theoretical approach

Given the results replicate the breadwinner norm, and demonstrate a significant variation across proxied wealth, I propose a model based on the work of Akerlof and Kranton (2000) in which identity enters utility directly. The crux of the model is a parameter through which a couple receives utility from adhering to a shared social identity, one component of which is that the husband out-earns the wife. Heterosexual couples, I argue, take on their own shared identity, which inherits norms from the wider society. A couple's behavior is often described through the complementary roles each partner plays. One can conjure a number of conversations in which a defiant wife—or a bitter friend—claims that *she wears the pants* in the marriage. The statement is not without remonstrance. It's implied the wife is acting in a leadership role against a well-established norm. The identity of a married couple is defined by each partner adhering to gender roles through their behavior within the couple.⁶

Both partners contribute to the shared identity and, in adhering to their role within the couple, receive joint utility. Once two individuals form a couple I assume they subscribe to prevailing social norms, one of which is that the husband serves as breadwinner. At minimum his role is to be career-oriented, which should be reflected in working and earning more. I define the breadwinner norm as the requirement that the husband earn at least as much as his wife, and treat it as a continuous threshold: $(w_f h_f - w_m h_m) \geq 0$ constitutes a violation and imposes disutility on the couple.⁷

⁶The strength of those norms is determined by general belief, or consensus, with the expectation that heterogeneity exists. The estimated model does not allow the norm to vary with demographic or geographic characteristics; Section B.6 sets out what such an extension would require and why it is not identified here.

⁷Individual identity, and the role of identity in occupational search and selection, lie outside the scope of

6.1. Model setup

I develop a static, one-period model where the unit of observation is the household. It is solved in partial equilibrium, with wages determined exogenously. It is a one-period model, so there is no saving. Two household members $i \in \{m, f\}$ have time endowment T . Only the spouses work, hours are $h_i = T - L_i$ with leisure L_i , and home production is not modeled. Wages are strictly positive and exogenously determined. We know $w_i > 0$, $y \geq 0$ such that non-labor income is never negative (no debt), and $C \geq 0$ is household consumption. Joint utility is additively separable,

$$(5) \quad U(L_m, L_f, C) = u(C) + v^m(L_m) + v^f(L_f),$$

with $u' > 0$, $u'' < 0$, $v^{i'} > 0$, $v^{i''} < 0$. Assuming a single decision maker, the household solves

$$(6) \quad \max_{L_m, L_f, C} u(C) + v^m(L_m) + v^f(L_f) - \alpha V(h_m, h_f)$$

subject to

$$\begin{aligned} C &= w_m h_m + w_f h_f + y - F \cdot (\mathbf{1}[h_m > 0] + \mathbf{1}[h_f > 0]), \\ h_i &= T - L_i, \quad h_i \geq 0, \quad L_i \geq 0, \quad C \geq 0. \end{aligned}$$

F is a fixed cost of entering the labor market, capturing commuting, childcare costs and the purchase of other services that may substitute for home production. It is expressed as a share of median household income in the year,

$$(7) \quad F = f \cdot \text{median}(y_t),$$

so that it requires no external calibration and rescales itself across a sample spanning four decades.

6.2. Identity framework

The norm applies only when gender roles are reversed. The model assigns no utility to compliance and imposes a penalty only on violation. Defining the earnings gap and the penalty as functions of hours,

$$(8) \quad \delta(h_m, h_f) \equiv w_f h_f - w_m h_m, \quad V(h_m, h_f) \equiv \max\{w_f h_f - w_m h_m, 0\},$$

the model. I isolate the norm channel so that it acts directly on the couple. The norm, I will show, is elastic in household consumption by derivation.

V is zero when the husband out-earns the wife and positive otherwise. Following Akerlof and Kranton (2000), violation generates a shared identity cost

$$(9) \quad I_j = -\alpha_j^{\text{eff}} V(h_m, h_f),$$

where the effective intensity of the norm is

$$(10) \quad \alpha_j^{\text{eff}} = \underbrace{\alpha_j(P_j)}_{\text{baseline}} \times \underbrace{g(\bar{\delta}_{-j})}_{\text{social enforcement}}, \quad g' > 0, \quad g(0) = 1,$$

with P_j the household's internalized prescription and $\bar{\delta}_{-j}$ the mean earnings gap in its reference group. To solve the model in estimation $g(\cdot) \equiv 1$, for reasons of endogeneity set out in Section B.6. $\hat{\alpha}$ is therefore the norm's effective intensity, inclusive of whatever enforcement operates.

6.3. Assumptions

Two functional-form assumptions allow the model to be taken to data. Household consumption utility is logarithmic,

$$(11) \quad u(C) = \log C,$$

which yields a wedge linear in consumption,

$$(12) \quad \tau_j = \frac{\alpha_j^{\text{eff}}}{u'(C)} = \alpha_j^{\text{eff}} \cdot C.$$

The disutility of work is iso-elastic,

$$(13) \quad v^i(T - h_i) = -\frac{\kappa}{1 + 1/\epsilon} h_i^{1+1/\epsilon},$$

where ϵ is the Frisch elasticity and κ is the marginal disutility of *market* work. Since home production is not modeled, κ should not be read as a taste for household production but instead it is the cost of supplying an hour to the market.

κ is imposed symmetric across spouses, and this is an important choice of restriction. While it may not seem the most accurate, assigning differential κ values would negate or explain away what the norm is meant to capture. All gender asymmetry in the model therefore arises from the observed wage gap and from α alone.

6.4. Optimality

Substituting the budget constraint, the household chooses hours to solve

$$(14) \quad \max_{h_m, h_f \geq 0} u(w_m h_m + w_f h_f + y - F(\mathbf{1}[h_m > 0] + \mathbf{1}[h_f > 0])) + v^m(T - h_m) + v^f(T - h_f) - \alpha_j^{\text{eff}} V(h_m, h_f).$$

This problem cannot be characterized by a single first-order condition. The objective is continuous but not differentiable where earnings are equal, because V is kinked there, and the fixed cost F makes the budget set non-convex at zero hours. The solution is therefore characterized regime by regime, and the household takes whichever attainable regime yields the highest utility. There are four potential solutions that are addressed.

Regime I: the norm is slack ($w_m h_m > w_f h_f$). The penalty is inactive, $V = 0$, and the problem reduces to the norm-free optimum with undistorted first-order conditions,

$$(15) \quad \frac{v^{m'}(L_m)}{u'(C)} = w_m, \quad \frac{v^{f'}(L_f)}{u'(C)} = w_f$$

Under the functional forms of Section 6.3 these become $\kappa h_i = w_i/C$ for $i \in \{m, f\}$.

Regime II: the norm binds ($w_f h_f > w_m h_m$). Here $V = w_f h_f - w_m h_m > 0$ and both conditions are distorted by a common wedge,

$$(16) \quad \frac{v^{m'}(L_m)}{u'(C)} = w_m(1 + \tau_j), \quad \frac{v^{f'}(L_f)}{u'(C)} = w_f(1 - \tau_j),$$

which under the assumed functional forms are $\kappa h_m = w_m(1 + \tau)/C$ and $\kappa h_f = w_f(1 - \tau)/C$. The wedge is therefore derived as

$$(17) \quad \tau_j \equiv \frac{\alpha_j^{\text{eff}}}{u'(C)} = \alpha_j^{\text{eff}} C, \quad \tilde{w}_m = w_m(1 + \tau_j), \quad \tilde{w}_f = w_f(1 - \tau_j),$$

the second equality following from the form of the log utility function. A single wedge governs both spouses because V is defined over one gap viewed from opposite sides. The norm acts as a proportional subsidy on the husband's labor and an equal proportional tax on the wife's, aligning with (Lee 2025). An interior solution for the wife requires $\tau_j < 1$; at $\tau_j \geq 1$ the implied tax exceeds her entire wage and the corner dominates. The estimates in Section 7.2 lie between 0.17 and 0.31, so the restriction never binds in practice, but it is a property of the solution rather than an assumption imposed on it.

Regime III: the kink ($w_f h_f = w_m h_m$). Neither interior condition need hold. Households for whom the Regime I optimum would violate the norm, but for whom the Regime II optimum overshoots, locate exactly at equal earnings. Because a continuum of households maps to a single point, this regime generates a mass point in the distribution of the wife's earnings share. This is the same bunching documented by Bertrand, Kamenica, and Pan (2015) and replicated in Section 5.1. It is also the reason the penalty is specified as continuous rather than as an indicator in this model. An indicator would simply imply that a discrete jump in behavior at the threshold occurs. This model allows for a continuous norm which allows for a kink to form.

Regime IV: the corner ($h_f = 0$). The wife does not work, the household pays the fixed cost once rather than twice, and $V = 0$ because $w_f \cdot 0 < w_m h_m$. This regime is governed by F , the fixed cost of entering the labor market, and by her wage, not by the norm. Therefore, withdrawal sets the penalty to zero, but so does bunching at the kink, and bunching preserves her earnings while withdrawal forfeits them. Regime III therefore dominates Regime IV for any norm-constrained household, at any α , a property returned to in Section 8.1.

TABLE 6. Regimes and first-order conditions

Regime	Condition	First-order conditions	Generates
I: norm slack	$w_m h_m > w_f h_f$	$\kappa h_i = \frac{w_i}{C}$	Undistorted hours
II: norm binds	$w_f h_f > w_m h_m$	$\kappa h_m = \frac{w_m(1+\tau)}{C}; \quad \kappa h_f = \frac{w_f(1-\tau)}{C}$	Reduced female hours
III: the kink	$w_f h_f = w_m h_m$	Neither holds; constrained optimum	Bunching at parity
IV: the corner	$h_f = 0$	$\kappa h_m = \frac{w_m}{C}, \quad V = 0$	Non-participation

6.5. Calibration

I then solve the static model using the same data, expanded to our largest set of observations in the sample. For every couple in the sample, 16,894,715 couples over 1980–2024, the model is solved. Separate estimates for each of the twenty seven year sample are provided. Because the data are repeated cross-sections, each year is a distinct t ; the fixed cost $F = f \cdot \text{median}(y_t)$ therefore rescales itself and requires no external calibration or deflation.

Of the number of women in married households in the sample, 26.5% are not working. This is data reported as having not worked in the last year. Two parameters are fixed and two are estimated. The Frisch elasticity is set to $\epsilon = 1$. The disutility parameter κ is inverted

from the Regime I first-order condition at sample means and is imposed symmetrically across both spouses. Table 7 summarizes the parameters outlined in the model and the choice variables.

In estimation, we set $g(\bar{\delta}_{-j}) \equiv 1$, so that $\alpha_j^{\text{eff}} = \alpha_t$. Here we collapse the function to one in order to avoid endogeneity. Discussed in more detail in Section 8.1, data that would exogenously estimate this equation was not available. The multiplier, as modeled here, is a function of compliance in the household's own reference group. For the sake of solving this model, it's important to note that compliance is itself an equilibrium outcome of the norm the multiplier is scaling as $\bar{\delta}_{-j}$ is determined jointly with the behavior it is meant to explain. Every household in a reference group moreover faces the same $\bar{\delta}$, so the group mean is collinear with group membership and α and g cannot be separately identified from a single cross-section (?). Estimating g would require exogenous variation in reference-group composition that this data does not provide, and attempting it without such variation overexerts the influence of $\hat{\alpha}$ in the model. Setting $g = 1$ therefore identifies α cleanly at the cost of absorbing any amplification into it, so $\hat{\alpha}$ should be read as the norm's effective intensity inclusive of whatever enforcement operates.

Exploratory simulation suggests the omission costs little in any case. At $\gamma = 4$ the multiplier amplifies α^{eff} by only 1.89 times, and no multiple equilibria arise for any γ up to eight. Since most couples satisfy the prescription simply because the husband earns more per hour, the analysis attributed 82% of compliance to the mechanics of the data. An extension is developed in Section B.6.

TABLE 7. Parameters

Parameter	Interpretation	Status	Source
ϵ	Frisch elasticity	Fixed at 1	Standard
κ	Disutility of market work	Calibrated	Regime I FOC at sample means
T	Time endowment	Fixed	4,000 annual hours
$g(\cdot)$	Social enforcement	Set to 1	Not identified in cross-section
α_t	Norm intensity	Estimated	Cliff ratio
f_t	Fixed cost, share of median income	Estimated	Corner share

Notes: α_t and f_t are estimated separately in each year against five moments, so the system is over-identified and the fit is testable. κ is symmetric across spouses by construction.

Solution. With log utility and $\epsilon = 1$ every regime reduces to a quadratic in C , so the model is solved in closed form: there is no value function to iterate, the problem being static, and no numerical optimization of any kind. Writing $A = w_m^2/\kappa$, $B = w_f^2/\kappa$ and $Y = y_0 - 2F$, Regime I solves $C^2 - YC - (A + B) = 0$ and Regime II solves $C^2 - [Y + \alpha(A - B)]C - (A + B) = 0$,

with analogous expressions at the kink and the corner; hours follow directly from the first-order conditions and the household takes the utility-maximizing regime. The closed form was verified against a dense two-dimensional grid search, which uses none of the algebra being tested.

7. Results

7.1. Income-elasticity of the norm parameter

From the solution of the model, we can state that norm parameter is income-elastic, where the norm parameter is an essential tax rate on a wife's wage. It is important to distinguish that the parameter derived in the model is not mathematically equivalent to the evidence from the regression analysis performed in Section 5.2. The tax is rising in consumption where wealth does not enter the functional form of the norm.

PROPOSITION 1. *The norm wedge is strictly increasing in household consumption: since $u'' < 0$,*

$$(18) \quad \frac{\partial \tau_j}{\partial C} = \frac{-\alpha_j^{\text{eff}} u''(C)}{[u'(C)]^2} > 0.$$

Richer households face a larger proportional distortion for the same α_j^{eff} , because the marginal-utility cost of forgoing the wife's earnings is lower. The norm is therefore more distortionary at higher resources. This results demonstrates that when the wedge parameter is derived, the penalty is a function of consumption as a result of the derivation. The penalty is unit-elastic under log utility, following along the shape of u . This wedge also holds for any value of u , so it does not explicitly depend on the logarithmic specialization adopted in Section 6.3.

7.2. Simulation results

Two moments are targeted to test the model's performance. Firstly, I want to re-generate the cliff ratio as reported in Bertrand, Kamenica, and Pan (2015), and the corner share which is an estimate of 1-*FLFP*. The latter can also be simulated within husband's-wage quintiles. Table 8 reports the fit averaged over all twenty-seven years.

Table 8 provides a comparison of the moments identified with the data and through our model. The generated moments from the model strongly corroborate behavior of the data. The un-targeted moments here serve as significant evidence to the strength of this simple model. A two-parameter model fitted to the cliff and the corner is under no obligation to reproduce the wife's share of couple hours to within 0.017, or the share of couples in which she out-earns her husband to within 0.014, in every year for four decades.

TABLE 8. Moment fit, averaged over 1980–2024

Moment	Data	Model	MAE	Status
Cliff ratio	1.575	1.592	0.019	Targeted
Corner share	0.255	0.248	0.008	Targeted
Corner share, Q1	0.277	0.140	0.137	Targeted
Corner share, Q5	0.301	0.396	0.095	Targeted
Wife's share of couple hours	0.398	0.382	0.017	Untargeted
Wife out-earns husband	0.258	0.245	0.014	Untargeted

Notes: MAE is the mean absolute deviation between model and data across the twenty-seven annual estimates. The un-targeted moments were never used in estimation in any year.

Model against data, every year 1980–2024

Top row: the two moments the model was fitted to. Bottom row: two moments held back entirely. The untargeted moments track the data to within 0.017 and 0.014 on average across 27 years.

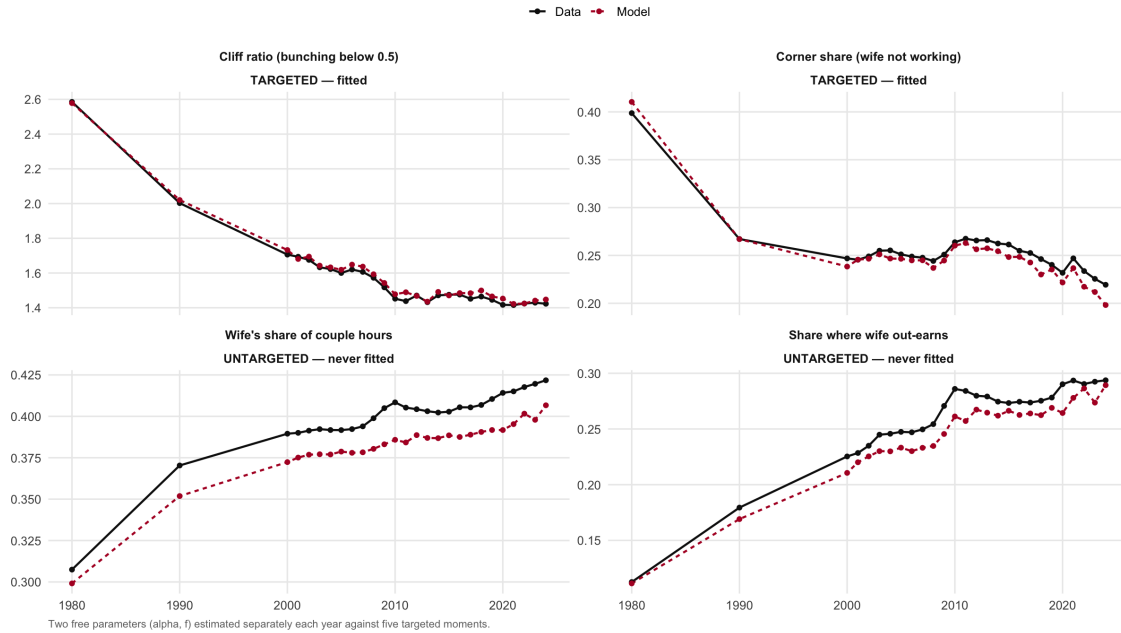


FIGURE 7. Model against data for 1980–2024

Notes: Top row, the two moments the model is fitted to; bottom row, two moments held back entirely. Decennial census years are marked separately from ACS years.

Next, Table 9 reports the estimates at selected years. α falls by 89 percent across the sample, as expected given the parameter must fall simply to hold the wedge constant as nominal incomes rise. Meaningfully, we can see the wedge does seem to be steadily declining, as progress on women's equality took hold. Among households where the norm actually binds falls from 0.262 to 0.172, a decline of 34 percent across our sample period. This results states that the norm that taxed the wife's marginal earnings at roughly 26% in 1980 taxes them at 17% today.

TABLE 9. Estimates and implications, selected years

	1980	1990	2000	2010	2020	2024
$\alpha (\times 10^6)$	11.4	5.61	3.74	2.42	1.95	1.20
f (share of median income)	0.109	0.080	0.086	0.099	0.081	0.077
τ , norm-binding households	0.262	0.236	0.241	0.204	0.220	0.172
Households the norm binds on (%)	15.9	21.8	26.4	27.6	30.0	30.3
Female hours lost (%)	6.12	6.22	7.28	7.01	7.51	6.02
Hours lost per affected household	361	347	380	341	358	296
Full-time equivalents lost (m)	1.17	1.70	2.54	2.57	2.91	2.47

Notes: α and f are estimated; the remaining rows are model implications. "Female hours lost" is the counterfactual of Section 7.3.

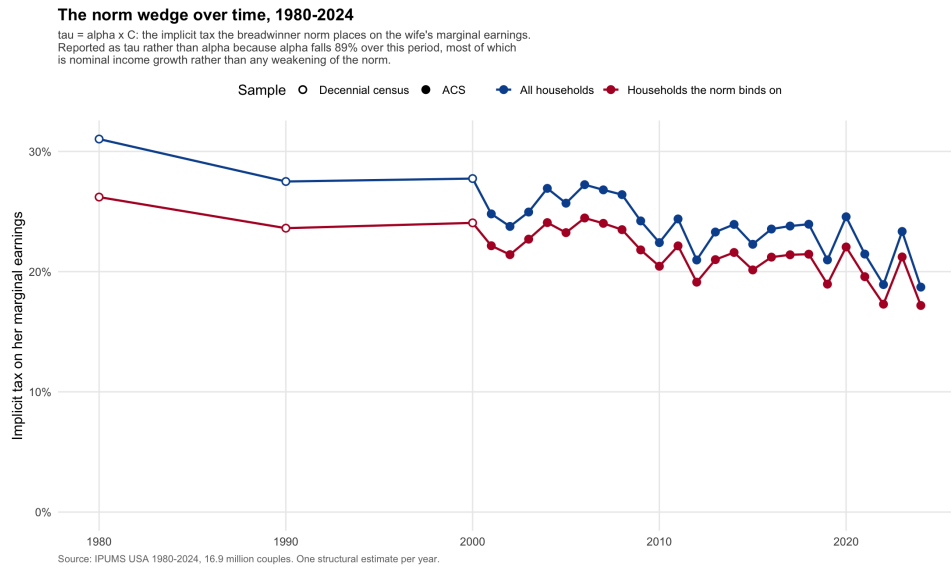


FIGURE 8. The norm wedge $\tau = \alpha C$, 1980–2024

7.3. Counterfactual

To conduct a counterfactual exercise, I choose to compare the norm's effect on lost labor hours against a world in which no norm exists. Complementary analysis may consider a world against less progress, and a higher social norm imposition, to estimate the gains from a reduction in α . For now, I create a counterfactual claim to estimate the size of losses each year due to the norm's existence. The model is re-solved with $\alpha = 0$, holding the fixed cost and preferences at their estimated values, so that only the norm is switched off. The difference in the wife's hours is the norm's own distortion. Because the corner share is invariant to α , the distortion is entirely on the intensive margin. If one were to remove the norm, the rate of entrance into the labor market should remain unchanged.

Counterfactual analysis finds the norm costs married women 6.12% of their market hours in 1980 and 6.02% in 2024. But that stability conceals two forces of comparable size running in opposite directions. Intensity per affected household falls, from 361 to 296 hours, while the share of households the norm binds on nearly doubles, from 15.9 to 30.3 percent, as wives' wages converge on their husbands' and far more couples cross into the region where the prescription is a live constraint rather than automatically satisfied. In levels the distortion more than doubles, from 1.17 to 2.47 million full-time equivalents, because population grows alongside exposure. So while the imposition of social norms on couples may feel less intense, as more women reach parity with their husbands in terms of income, the number of lost hours worked rises.

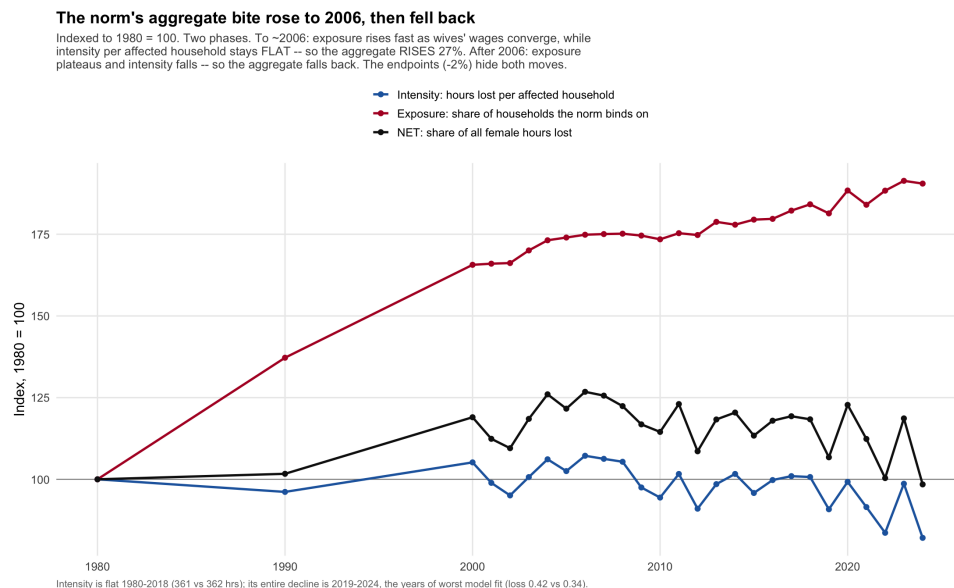


FIGURE 9. Intensity and exposure, indexed to 1980

A more interesting framing of this compositional analysis is that the expansion in exposure is not uniquely a story about women. The norm's threshold is the husband's

earnings, and his real median wage rose 8% over the period, having been flat from 1980 to 2015. Setting his wages to grow at the rate women's did and re-solving, the norm would bind on 18.5% of couples in 2024 rather than 30.3%. The model is successful in providing enough flexibility on how the norm is imposed on a couple to see how the lack of growth in wages for married men can seep into their partners' decision to work. Nearly 80% of the growth in in couples nearing the 50/50 income threshold is not attributable to women making significant gains in their wages, but to male wage stagnation.

Recall the wedge is a proportional subsidy on the husband's labor and an equal proportional tax on the wife's, and since he earns more per hour the subsidy moves more hours than the tax removes. In 2024 he gains 5.50 billion hours while she loses 4.94 billion. Households are nonetheless strictly worse off, being distorted away from their optimum.

8. Discussion

8.1. Limitations

Despite the model's strong fit to the data, there remain two core limitations that are structural rather than incidental. Firstly, the model is narrow in its scope of the intensive margin. Despite the failure to touch on both margins, the model performs well to describe a relative-earnings norm is an hours mechanism. This is a property of the prescription rather than a defect in fitting. At the kink $w_f h_f = w_m h_m$ exactly, so $V = 0$ and the penalty vanishes *while the wife retains her earnings*; withdrawing also sets $V = 0$ but costs her everything. Bunching therefore strictly dominates the corner at every α . The simulated corner share is invariant at 26.0% for α from zero to unity with no solutions in a corner.

Secondly, The model assigns non-participation to the wrong households. The aggregate corner share is matched almost exactly, but its distribution is not accurate to the types of households that have non-working wives. Across husband's-wage quintiles the model runs from 0.140 at Q1 to 0.396 at Q5, against data that run from 0.277 to 0.301. In the data the husband's wage barely predicts whether his wife works, whereas in the model it nearly determines it. This, of course, could be due to the lack of dynamics and a narrow approach to isolating just the mechanism of social norm imposition on working hours. Two candidate explanations were tested and rejected. Selection in the wage equation does not account for it. An attempt to shift imputed wages for non-workers down by 50% moves Q1 only from 0.148 to 0.164. The norm parameter appears robust to selection in wage. And similarly, preference heterogeneity cannot explain the mismatch either. Log utility's income effect appears to be very strong and governing a large portion of this assignment.

While this model is successful in fitting data for the object of identification, intensive margin work of married women, there are certain features that fail. Similar to the mismatch in the persistent issue of real business cycle models reported the wage-hours

correlation they could not match, I present a well-fitted model with recommendations for extensions and a discussion of its limitation. Model fit is also weakest at both ends of the sample, so the most recent estimates are the least reliable.

The model's known limitation: it puts non-participation in the wrong households

Share of wives not working, by the husband's wage quintile. The model matches the AGGREGATE corner share almost exactly, but assigns it by the income effect — too few non-workers among low-earning husbands, too many among high-earning ones. In the data his wage barely predicts it.

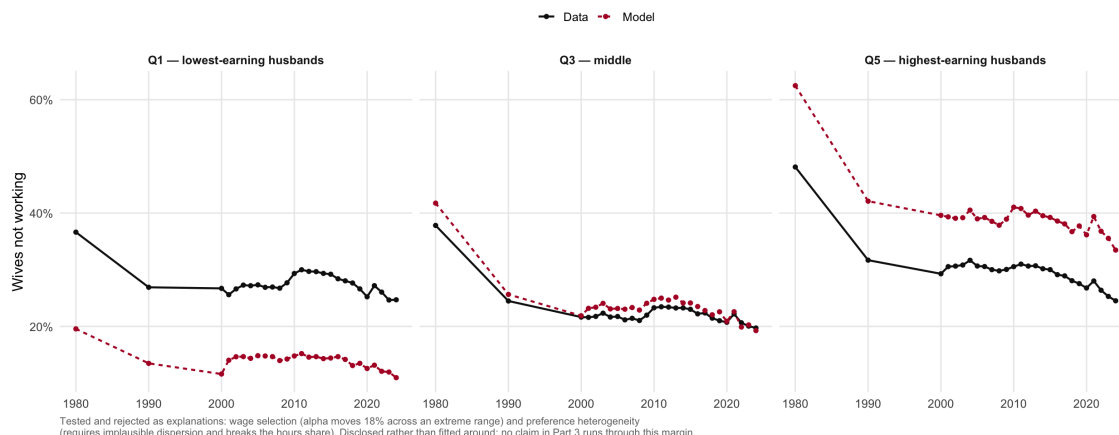


FIGURE 10. Where the model fails: non-participation by husband's wage quintile

9. Conclusion

Summary. This paper asks whether the breadwinner norm documented by Bertrand, Kamenica, and Pan (2015) has weakened, and finds that the question is ill-posed as usually stated. The norm has two margins and they have moved differently. Re-estimating BKP's own specification on the years since their sample ends, the effect of a wife's likelihood of out-earning her husband on *whether she works* falls by 37%, while its effect on *how much she earns relative to her capacity*, conditional on working, is essentially unchanged. A single declining parameter does not generate that pattern, since uniform weakening attenuates both margins together.

Separating the mechanisms requires distinguishing two accounts that produce the same reduced-form pattern. A standard wealth effect on the secondary earner predicts that richer households supply less female labor and so does an identity penalty that binds harder when the household can afford to indulge it. The distinguishing prediction is the interaction that a pure wealth effect shifts female labor supply smoothly and is indifferent to where the couple sits in the relative-earnings distribution, whereas an identity penalty predicts that the discontinuity at parity itself varies with household resources. The empirical analysis accordingly measures the gradient by interacting a housing-wealth proxy with cultural context rather than by regressing participation on wealth directly.

Implications. The model estimates deliver a result that neither the reduced-form evidence nor the raw parameter would reveal. The norm's intensity has fallen: among the households it binds on, the implicit tax on the wife's marginal earnings declines from 26 to 17% between 1980 and 2024. This confirms the evidence in the macroeconomic literature that women have seen enormous gains in terms of equality. But as the gains in equality occur, the reach of this specific breadwinner norm, however, has nearly doubled. Today 30% of couples have wives' wages converging on their husbands', introducing a new set of questions for couples who begin to cross this threshold.

The two forces are of comparable magnitude and largely offset: measured against a counterfactual in which the norm is switched off, it costs married women 6.1 percent of their market hours in 1980 and 6.0 percent in 2024, having peaked at 7.8 percent in 2006. In levels the distortion more than doubles, from 1.2 to 2.5 million full-time equivalents.

The expansion in reach is moreover not principally a story about women. The threshold the norm polices is the husband's earnings, and his real median wage rose eight percent over the period, having been flat from 1980 to 2015. The breadwinner norm binds on more couples today not because it strengthened, but because men stopped automatically out-earning their wives. This has many implications for future research on the behavioral responses of gender norms on both men and women, individually and as a couple.

References

- Acemoglu, Daron, David H. Autor, and David Lyle. 2004. "Women, War, and Wages: The Effect of Female Labor Supply on the Wage Structure at Midcentury." *Journal of Political Economy* 112 (3): 497–551.
- Akerlof, George A. and Rachel E. Kranton. 2000. "Economics and Identity." *The Quarterly Journal of Economics* 115 (3): 715–753.
- Arellano-Bover, Jaime, Nicola Bianchi, Salvatore Lattanzio, and Matteo Paradisi. 2025. "One Cohort at a Time: A New Perspective on the Declining Gender Pay Gap."
- Attanasio, Orazio, Erik Hurst, and Luigi Pistaferri. 2013. "The Evolution of Income, Consumption, and Leisure Inequality in the United States, 1980–2010." , National Bureau of Economic Research.
- Bertrand, Marianne, Emir Kamenica, and Jessica Pan. 2015. "Gender Identity and Relative Income within Households." *The Quarterly Journal of Economics* 130 (2): 571–614.
- Binder, Ariel J. and David Lam. 2018. "Is There a Male Breadwinner Norm? The Hazards of Inferring Preferences from Marriage Market Outcomes." NBER Working Paper 24907, National Bureau of Economic Research. <https://www.nber.org/papers/w24907>. Revised 2019.
- Blundell, Richard, Luigi Pistaferri, and Itay Saporta-Eksten. 2016. "Consumption Inequality and Family Labor Supply." *American Economic Review* 106 (2): 387–435.
- Bursztyn, Leonardo, Alessandra L. González, and David Yanagizawa-Drott. 2020. "Misperceived Social Norms: Women Working Outside the Home in Saudi Arabia." *American Economic Review* 110 (10): 2997–3029. 10.1257/aer.20180975.
- Cesarini, David, Erik Lindqvist, Matthew J. Notowidigdo, and Robert Östling. 2017. "The Effect of Wealth on Individual and Household Labor Supply: Evidence from Swedish Lotteries." *American Economic Review* 107 (12): 3917–3946.
- Chiappori, Pierre-André. 1992. "Collective Labor Supply and Welfare." *Journal of Political Economy* 100 (3): 437–467.
- Disney, Richard and John Gathergood. 2018. "House Prices, Wealth Effects and Labour Supply." *Economica* 85 (339): 449–478. 10.1111/ecca.12253.
- Gethin, Amory and Emmanuel Saez. 2025. "Global working hours." , NBER. 10.3386/w34217.
- Goldin, Claudia. 1990. *Understanding the Gender Gap: An Economic History of American Women*. Oxford University Press.
- Goldin, Claudia and Lawrence F. Katz. 2002. "The Power of the Pill: Oral Contraceptives and Women's Career and Marriage Decisions." *Journal of Political Economy* 110 (4): 730–770.
- Goldin, Claudia, Sari Pekkala Kerr, and Claudia Olivetti. 2022. "When the Kids Grow Up: Women's Employment and Earnings across the Family Cycle." NBER Working Paper 30323, National Bureau of Economic Research.
- Goldin, Claudia and Joshua Mitchell. 2016. "The New Life Cycle of Women's Employment: Disappearing Humps, Sagging Middles, Expanding Tops." NBER Working Paper 22913, National Bureau of Economic Research. Was titled "Women's Work and the Gender Wage Gap" (BPEA) — that title does not match the PDF in /literature. Corrected to the paper you hold. Also published JEP 31(1), 2017, pp. 161–182 — pick one.
- Greenwood, Jeremy, Ananth Seshadri, and Mehmet Yorukoglu. 2005. "Engines of Liberation." *The Review of Economic Studies* 72 (1): 109–133.

- International Labour Organization and World Bank. 2026. "Ratio of Female to Male Labor Force Participation Rates (ILO)." ILOEST/ILOSTAT and World Bank staff estimates, World Development Indicators 129. Processed by Our World in Data. <https://archive.ourworldindata.org/20260806-091206/grapher/ratio-of-female-to-male-labor-force-participation-rates-ilo-wdi.html>. Archived 6 August 2026; retrieved 31 August 2026.
- Kleven, Henrik and Camille Landais. 2017. "Gender Inequality and Economic Development: Fertility, Education and Norms." *Economica* 84 (334): 180–209. <https://doi.org/10.1111/ecca.12230>.
- Ko, Paul I., Gee Young Oh, and Sally Zhang. 2024. "Gender Norms, Intrahousehold Bargaining, and the Dynamics of Female Labor Supply." <https://events.bse.eu/live/files/5653-normsbargaininglaborsupplydec2024pdf>. Working paper.
- Lee, Jay Euijung. 2025. "Marriage and Misallocation: Evidence from 70 Years of U.S. History." CEP Discussion Paper 2119, Centre for Economic Performance, LSE. https://cep.lse.ac.uk/_NEW/PUBLICATIONS/abstract.asp?index=11726.
- Lundberg, Shelly. 1985. "The Added Worker Effect." *Journal of Labor Economics* 3 (1): 11–37.
- Melvin Stephens, Jr. 2002. "Worker Displacement and the Added Worker Effect." *Journal of Labor Economics* 20 (3): 504–537.
- National Center for Health Statistics. 2025. "Births in the United States, 2024." NCHS Data Brief No. 535, Centers for Disease Control and Prevention.
- Olivetti, Claudia. 2013. "The female labor force and long-run development: the American experience in comparative perspective." , National Bureau of Economic Research.
- Pew Research Center. 2022. "Young Women Are Out-Earning Young Men in Several U.S. Cities." Pew Research Center report.
- Ramey, Valerie A. and Neville Francis. 2009. "A Century of Work and Leisure." *American Economic Journal: Macroeconomics* 1 (2): 189–224.
- Rock, Bram De, Mariia Kovaleva, and Tom Potoms. 2025. "Housing Wealth, Marital Stability and Labor Supply: An Intertemporal Analysis." IFS Working Paper 25/13, Institute for Fiscal Studies. <https://ifs.org.uk/publications/housing-wealth-marital-stability-and-labor-supply-intertemporal-analysis>.
- Rose, Evan K. 2018. "The Rise and Fall of Female Labor Force Participation During World War II in the United States." *Journal of Economic History*. TODO verify volume/pages.
- Rosenberg, Sarah. 2024. "Revisiting the Breadwinner Norm: Replicating Bertrand, Kamenica, and Pan (2015)." Working Paper 2024:10, Lund University, Department of Economics. https://ideas.repec.org/p/hhs/lunewp/2024_010.html. TODO verify author first name.
- Sakamoto, Tomoko and Miki Kohara. 2025. "Why Gender Norms Matter." *Economica*. TODO verify author names, volume and issue.
- Stansbury, Anna and Lawrence H Summers. 2020. "The Declining Worker Power Hypothesis: An explanation for the recent evolution of the American economy." Working Paper 27193, National Bureau of Economic Research. 10.3386/w27193.
- U.S. Bureau of Labor Statistics. 2026. "Labor Force Participation Rate — Men [LNS11300001]." Retrieved from FRED, Federal Reserve Bank of St. Louis. <https://fred.stlouisfed.org/series/LNS11300001>. Accessed 31 August 2026.

- Wickens, Michael. 2012. *Macroeconomic Theory: A Dynamic General Equilibrium Approach Second Edition*, vol. None of *Economics Books*. 1 ed. Princeton University Press. None.
- Wu, Alice H. 2020. “Gender Bias among Professionals: An Identity-Based Interpretation.” *The Review of Economics and Statistics* 102 (5): 867–880.
- Zinovyeva, Natalia and Maryna Tverdostup. 2021. “Gender Identity, Coworking Spouses, and Relative Income within Households.” *American Economic Journal: Applied Economics* 13 (4): 258–284. TODO verify pages.

Appendix A. Data appendix

The ACS survey selection was taken at the decennial where one-year survey was not available. This covered the years 1970, 1980, and 1990. Beginning in 2000, one-year ACS surveys were taken. This is due to a choice to choose the same survey sampling technique for consistency and use the most recently released data. While three-year data was available for the periods between 2001 and 2013, the US Census Bureau stopped releasing three-year surveys. As such, one-year surveys were used in favor of five-year. Given geographies would not play an important role in the analysis, one-year surveys are both acceptable and consistent across a majority of the study period.

Appendix B. Proofs and comparative statics

This appendix derives the first- and second-order conditions summarized in Table 6, proves Proposition 1, gives the closed-form solution used by the solver, and reports the comparative statics evaluated at the estimated parameters.

B.1. Setup and notation

Substituting the budget constraint into (6), write the household objective as a function of hours alone:

$$(A1) \quad W(h_m, h_f) = u(C(h_m, h_f)) - \frac{\kappa}{1+1/\epsilon} (h_m^{1+1/\epsilon} + h_f^{1+1/\epsilon}) - \alpha^{\text{eff}} V(h_m, h_f),$$

$$C(h_m, h_f) = w_m h_m + w_f h_f + y - F(\mathbf{1}[h_m > 0] + \mathbf{1}[h_f > 0]), \quad V(h_m, h_f) = \max\{w_f h_f - w_m h_m, 0\}.$$

Since $v^i(L_i) = -\frac{\kappa}{1+1/\epsilon} h_i^{1+1/\epsilon}$ with $h_i = T - L_i$, the marginal disutility of leisure is $v^{i'}(L_i) = \kappa h_i^{1/\epsilon}$, which is the object appearing on the left of (15) and (16). Throughout, $\alpha \equiv \alpha^{\text{eff}}$, since $g(\cdot) \equiv 1$ in estimation.

Two structural facts organize the derivation.

LEMMA A1 (Concavity of the interior problem). *On the open region $\mathcal{H} = \{(h_m, h_f) : h_m > 0, h_f > 0\}$, where the number of fixed costs paid is constant at two, W is strictly concave.*

PROOF. C is affine in (h_m, h_f) on $\mathcal{H}$ and u is concave and increasing, so $u(C(\cdot))$ is concave. Each disutility term is strictly concave in h_i for $\epsilon > 0$. V is the maximum of two affine functions and hence convex, so $-\alpha V$ is concave for $\alpha \geq 0$. A sum of concave functions, at least one strictly so, is strictly concave. $\square$

Lemma A1 is worth stating explicitly because it locates the model's only genuine non-convexity. The identity penalty does *not* introduce one: V is kinked but convex, so $-\alpha V$

TABLE A1. Sample construction by year

Year	Person records	Women	Aged 18–65	Spouse linked	Mutual pair	Husband working
1970	4,060,019	2,088,154	1,181,732	812,075	768,399	704,084
1980	11,343,120	5,844,944	3,559,394	2,239,556	2,113,655	1,849,245
1990	12,501,046	6,443,186	3,936,090	2,520,130	2,372,264	2,058,037
2000	14,081,466	7,211,710	4,396,928	2,741,835	2,565,877	2,146,354
2001	1,192,206	618,464	381,531	246,166	229,576	192,855
2002	1,074,628	557,760	344,456	221,937	206,913	171,741
2003	1,194,928	619,100	383,220	246,424	228,438	189,532
2004	1,194,354	619,411	383,227	246,588	228,324	190,100
2005	2,878,380	1,490,317	921,938	590,574	545,634	453,405
2006	2,969,741	1,526,291	945,667	592,255	546,366	464,496
2007	2,994,662	1,539,907	955,346	597,575	552,777	467,444
2008	3,000,657	1,544,657	957,318	593,842	549,908	472,688
2009	3,030,728	1,559,401	965,954	593,905	548,819	453,751
2010	3,061,692	1,572,720	974,455	590,396	544,567	444,704
2011	3,112,017	1,593,519	985,473	573,573	526,618	430,898
2012	3,113,030	1,596,079	987,809	577,039	528,947	437,488
2013	3,132,795	1,604,061	992,321	580,795	530,540	442,491
2014	3,132,610	1,603,381	986,434	574,740	522,438	440,001
2015	3,147,005	1,610,169	986,972	575,602	522,055	442,964
2016	3,156,487	1,612,346	983,904	573,507	518,620	442,110
2017	3,190,040	1,628,854	992,887	579,515	523,680	449,724
2018	3,214,539	1,639,921	994,978	580,527	522,504	451,230
2019	3,239,553	1,651,221	993,248	583,644	523,719	454,363
2020	2,641,054	1,335,408	800,356	462,162	414,128	352,507
2021	3,252,599	1,654,935	983,030	574,327	513,232	436,563
2022	3,373,378	1,717,163	1,015,936	593,563	530,264	459,177
2023	3,405,809	1,733,894	1,018,717	594,000	528,980	460,000
2024	3,422,888	1,743,242	1,017,171	590,464	525,455	457,695
Total	109,111,431	55,960,215	34,026,492	20,846,716	19,232,697	16,415,647

Notes: Unweighted counts. *Person records* is the full extract for that sample. *Women* and *aged 18–65* are successive restrictions on individuals; *spouse linked* counts women with a spouse pointer; *mutual pair* additionally requires each partner's pointer to identify the other, which is the couple definition used throughout; *husband working* applies the further restriction imposed by Bertrand, Kamenica, and Pan (2015) in their labor supply specifications. Rows above the gap are decennial censuses (1970, 1980 and 1990 at 1 percent, 2000 at 5 percent); rows are ACS single-year files, which expand from roughly 1.2 million to 3 million person records when the ACS reached full sample size in 2005. The 2020 file is smaller because of pandemic-related collection disruption.

preserves concavity and merely renders W non-differentiable on the ray $w_f h_f = w_m h_m$. The non-convexity is the fixed cost F , which makes W jump upward at $h_f = 0$ as one cost is refunded. Consequently Regimes I–III are the three Karush–Kuhn–Tucker cases of a single concave program and are pinned down by first-order conditions, whereas Regime IV must be selected by a discrete utility comparison rather than by any marginal condition.

B.2. First-order conditions

Interior regimes. On $\mathcal{H}$, W is differentiable except where $\delta \equiv w_f h_f - w_m h_m = 0$. A generalized gradient is obtained from $\partial \max\{\delta, 0\} = \{\lambda\}$ where we can define λ such that:

$$(A2) \quad \lambda = \begin{cases} 0 & \text{if } \delta < 0 \quad (\text{Regime I}), \\ [0, 1] & \text{if } \delta = 0 \quad (\text{Regime III}), \\ 1 & \text{if } \delta > 0 \quad (\text{Regime II}). \end{cases}$$

Since $\partial \delta / \partial h_f = w_f$ and $\partial \delta / \partial h_m = -w_m$, stationarity of (A1) requires that there exist a λ consistent with (A2) such that

$$(A3) \quad 0 = u'(C) w_f - \kappa h_f^{1/\epsilon} - \lambda \alpha w_f,$$

$$(A4) \quad 0 = u'(C) w_m - \kappa h_m^{1/\epsilon} + \lambda \alpha w_m.$$

Dividing through by $u'(C) > 0$ and collecting terms,

$$(A5) \quad \frac{v^{f'}(L_f)}{u'(C)} = w_f (1 - \lambda \tau), \quad \frac{v^{m'}(L_m)}{u'(C)} = w_m (1 + \lambda \tau), \quad \tau \equiv \frac{\alpha}{u'(C)}.$$

Equation (A5) is the interior optimum, and the three interior regimes are the three admissible values of λ . Setting $\lambda = 0$ returns (15); setting $\lambda = 1$ returns (16), which is the derivation used in Section 6. The wedge enters the two conditions with opposite signs and equal magnitude because V depends on only the spousal hours only through δ , so one unit of the wife's earnings relaxes the constraint by exactly as much as one unit of the husband's tightens it. The norm is a proportional tax of τ on her wage and a proportional subsidy of τ on his. An interior solution for the wife requires $\lambda \tau < 1$ otherwise (A3) has no positive root and the corner dominates.

The kink. Regime III is the intermediate case $\lambda \in (0, 1)$, subject to the additional restriction $w_f h_f = w_m h_m$. A household bunches exactly when the two differentiable optima fall on the wrong sides of the threshold: the $\lambda = 0$ solution violates the norm ($\delta > 0$) while the $\lambda = 1$ solution overshoots ($\delta < 0$). The set of (w_m, w_f, y) satisfying both inequalities has positive measure, which is why a continuum of households maps to a single point and the

earnings-share density has a mass point at one half. Bunching is therefore a prediction of the continuous penalty. If it was a simple threshold or an indicator penalty, instead it would make W discontinuous at $\delta = 0$, resulting in a jump.

The corner. Regime IV is evaluated comparatively, With $h_f = 0$ the household pays F once, $V = 0$ identically, and the husband solves $\kappa h_m^{1/\epsilon} = u'(C)w_m$ with $C = w_m h_m + y - F$. The wife withdraws when the resulting W exceeds the best interior value.

B.3. Second-order conditions

Within a differentiable regime, V is eliminated from the Hessian of (A1), which is

$$(A6) \quad H = \begin{pmatrix} u''(C) w_m^2 - \frac{\kappa}{\epsilon} h_m^{1/\epsilon-1} & u''(C) w_m w_f \\ u''(C) w_m w_f & u''(C) w_f^2 - \frac{\kappa}{\epsilon} h_f^{1/\epsilon-1} \end{pmatrix}.$$

Write $q_i \equiv (\kappa/\epsilon) h_i^{1/\epsilon-1} > 0$. The diagonal entries are $u'' w_i^2 - q_i < 0$ since $u'' < 0$ (decreasing returns to marginal utility).

B.4. Proof of Proposition 1

From (17), $\tau = \alpha/u'(C)$. Differentiating with respect to C , treating α as a household primitive,

$$(A7) \quad \frac{\partial \tau}{\partial C} = \alpha \frac{d}{dC} \left[\frac{1}{u'(C)} \right] = \frac{-\alpha u''(C)}{[u'(C)]^2} > 0,$$

since $\alpha > 0$, $u'' < 0$, and $[u']^2 > 0$. This is (18). Expressed as an elasticity we can see the results more clearly such that:

$$(A8) \quad \frac{\partial \ln \tau}{\partial \ln C} = \frac{-C u''(C)}{u'(C)} = \gamma(C),$$

the coefficient of relative risk aversion. The elasticity of the norm wedge, τ , with respect to household consumption is equivalent to the curve of utility derived from consumption. Under the logarithmic specification $\gamma \equiv 1$, so $\tau = \alpha C$ is unit-elastic: a household with twice the consumption faces twice the proportional distortion from the same α . Nothing in (A7) uses the functional forms of Section 6.3.

B.5. Comparative statics

Table A2 perturbs each primitive by ten percent at the parameters fitted to 2019, holding the wage distribution fixed, and reports the resulting percentage change in five moments.

TABLE A2. Comparative statics at the fitted 2019 parameters

+10% change in	% change in outcome				
	Wife hrs	Husb. hrs	Wife earn. share	Cliff	Corner share
α (norm intensity)	-0.55	+0.42	-0.51	+2.80	+0.02
w_m (husband's wage)	-6.10	+2.78	-4.97	-49.25	+9.71
w_f (wife's wage)	+6.07	-2.75	+4.86	+102.23	-17.80
y (nonlabor income)	-0.19	-0.16	-0.03	+0.19	+0.29
F (fixed cost)	-0.88	+0.44	-0.68	-0.15	+9.39
κ (work disutility)	-4.88	-4.70	-0.10	-1.21	+4.79

Notes: Each row raises one primitive by ten percent for every household, re-solves the model exactly, and reports the percentage change in the weighted moment. “Cliff” is the ratio of density just below to just above an equal earnings split as it is referred to in the replication paper. “Corner share” is the fraction of wives at $h_f = 0$.

Incidence. Table A3 reports a within-household counterfactual such that each couple is solved twice, once at its estimated α and once at $\alpha = 0$, and the difference in the wife's annual hours is the norm's cost for that household. This is a decomposition of the estimated model, not a comparison across couples, so it is free of any composition effects.

The incidence is sharply non-monotone in the husband's wage, and in a way the reduced-form evidence cannot show. Reach falls steeply across his quintiles, from 42 percent in Q1 to 7 percent in Q5, because a high-earning husband puts the threshold out of his wife's reach and the norm is automatically satisfied. Intensity moves the opposite way: among couples where the norm does bind, the wife loses 238 hours a year in Q1 and 484 in Q5, roughly double.

B.6. Model extension

Three extensions are discussed in the construction of the model. Each is stated with what it would add and what identification it would require, since in every case the obstacle is data rather than tractability.

Endogenous social enforcement. Restoring $g(\bar{\delta}_{-j})$ with $g' > 0$ turns the model into an equilibrium object. Households no longer solve independently and the norm's strength depends on how many others comply, and their compliance depends on its strength, so

TABLE A3. Who bears the norm

	% of couples	% bound	Hours lost	Hours lost bound
<i>Panel A: by the couple's wage ratio w_f/w_m</i>				
< 0.50	28.4	0.0	0	—
0.50–0.75	19.4	0.0	0	—
0.75–1.00	17.4	0.0	0	—
1.00–1.25	10.8	99.3	201	201
> 1.25	24.1	76.7	354	404
<i>Panel B: by the husband's wage quintile</i>				
Q1 (lowest)	20	42.2	151	238
Q2	20	43.9	136	309
Q3	20	30.6	117	382
Q4	20	19.1	88	458
Q5 (highest)	20	7.2	35	484

Notes: “% bound” is the share of couples in the cell for which the norm alters the solution, that is, whose optimum lies in Regime II or III. “Hours lost” is the average reduction in a wife’s annual hours, weighted relative to the same household solved at $\alpha = 0$. This is then taken over all couples in the cell and the final column conditions on the bound. Panel B cells are quintiles by construction. Source:

the model must be solved as a fixed point in aggregate behavior,

$$\text{find } \bar{\delta}^* \text{ such that } \bar{\delta}(\alpha \cdot g(\bar{\delta}^*)) = \bar{\delta}^*.$$

Identification is essentially the binding constraint against the introduction of this extension. Separating α from g requires variation in reference-group composition that is not itself a response to the norm. Future analysis could include an assessment of appropriate proxies for the reference group.

Intra-household bargaining. This model is static, and as a single objective it is maximized and spouses do not bargain within the household. A collective formulation (Chiappori 1992) would replace it with a weighted sum, $\mu U^m + (1 - \mu) U^f$, where the Pareto weight μ represents bargaining power. The interesting version for identity economics is not to add the norm alongside bargaining but to let the norm operate *through* μ : a wife who out-earns her husband violates a prescription and, in doing so, alters the weight her preferences receive. That formulation makes the norm a determinant of who gets what within the household rather than only of how much each works, which is closer to what Akerlof and Kranton’s identity framework implies for couples. Ko, Oh, and Zhang (2024) take the collective route with a threshold norm; the present model gains a functional form disciplined by the observed kink and parameters estimated on microdata, and gives up the bargaining margin entirely.

Appendix C. Additional results and robustness

C.1. Replication tables

C.2. Robustness for empirical analysis

TABLE A4. Updated Sample: 1970–2024

	Wife LFP	Income Gap
ln(Husband Income)	0.0198*** (0.0071)	-0.2755*** (0.0182)
Cubic ln(Husband Income)	-0.1089*** (0.0103)	-0.1102*** (0.0315)
Cubic ln(Husband Income) + Any Children	-0.1167*** (0.0103)	-0.1187*** (0.0316)
Observations	249,998	165,402
Full sample	15,018,678	15,018,678

Standard errors in parentheses.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Preferred specification: cubic ln(Husband Income).

TABLE A5. Balance of observable characteristics by political orientation and wealth status

Variable	Conservative		Progressive	
	Not Wealthy	Wealthy	Not Wealthy	Wealthy
Female age	42.64	47.44	42.25	47.14
Male age	44.59	49.41	44.31	49.16
Female college	0.324	0.627	0.372	0.718
Male college	0.292	0.656	0.337	0.731
Number of children	1.22	1.34	1.24	1.32
Male labor income	53,882	129,263	56,274	147,758
Female labor income	28,852	49,107	31,893	62,815
Home value	201,132	593,215	263,158	878,028
Female LFP	0.721	0.687	0.723	0.719
Female hours	27.70	26.74	27.48	28.10

Notes: Entries report group means. Wealth status is defined according to the wealth classification of home ownership within-state top quartile. Female LFP denotes female labor force participation, and female hours denotes weekly hours worked.

TABLE A7. Estimated conservative–wealth interaction by year

Year	Interaction Estimate	Std. Error	<i>p</i> -value
2012	-0.0258	0.0078	0.0011
2013	-0.0209	0.0071	0.0035
2014	-0.0250	0.0090	0.0054
2015	-0.0363	0.0080	<0.001
2016	-0.0197	0.0072	0.0065
2017	-0.0306	0.0084	<0.001
2018	-0.0287	0.0082	<0.001
2019	-0.0282	0.0074	<0.001
2020	-0.0256	0.0083	0.0022

Notes: Entries report the estimated coefficient on the interaction between conservative political orientation and wealth status from separate regressions by year.

TABLE A8. Wife's labor supply by housing tier

Tier	Couples	LFP	Weekly hrs
Renter	715,502	68.66%	25.83
Owner, mortgaged	1,604,202	76.07%	29.42
Owner outright	336,609	64.16%	24.33
Owner outright, top-quartile value	99,065	60.07%	22.99

Notes: Household-weighted means, unconditional and without covariates. The sixteen-point spread between mortgaged owners and debt-free owners of high-value homes is the same leverage/wealth distinction that appears in the regression estimates of Table A10. Outright owners are older by construction.

TABLE A9. The culture $\times$ wealth interaction, by the wife's education

Wife's education	(1) High school or less	(2) Some college	(3) College or more
Republican-majority county	+1.012* (0.498)	-0.428 (0.370)	+0.726** (0.278)
Wealthy: top-quartile home value	+2.912*** (0.436)	+0.217 (0.457)	+2.835*** (0.345)
Republican $\times$ top-quartile home	-2.840*** (0.762)	-1.553* (0.620)	-2.883*** (0.548)
Demographic controls	Yes	Yes	Yes
State FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Husband income decile FE	Yes	Yes	Yes
Observations	777,287	561,366	1,109,930

Notes: The dependent variable is an indicator for the wife's labor force participation; coefficients are in percentage points, with standard errors in parentheses clustered on county. Each column estimates the full specification separately within one education group, so the estimates are not constrained to share any coefficient across columns. Demographic controls are both spouses' ages (wife's entered quadratically), the husband's college attainment, the number of own children (entered quadratically) and log husband's labor income; the wife's own college indicator is absorbed within each stratum. Observations do not sum to the pooled sample because education is unreported for a small number of wives. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

Interpretation: the interaction is negative and statistically significant in all three education groups, and is of almost identical magnitude at the two ends of the distribution (-2.84 for high school or less, -2.88 for college or more). The result is therefore not an education effect in disguise.

TABLE A10. The interaction across four wealth measures

Outcome	(1)	(2)	(3)	(4)
Wealth measured as	LFP Top-qtile home	Weekly hrs Top-qtile home	LFP Owns outright	LFP Asset income
Republican-majority county	+0.88*	+0.47***	+0.71*	+0.78*
Wealth main effect	+2.75***	+1.70***	-5.71***	-0.54**
Republican $\times$ wealth	-2.79***	-1.49***	-1.64***	-2.26***
Demographic controls	Yes	Yes	Yes	Yes
State FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Husband income decile FE	Yes	Yes	Yes	Yes
Observations	2,477,474	2,477,474	2,477,474	2,477,474

Notes: Columns 1, 3 and 4 report effects on the wife's labor force participation in percentage points; column 2 reports the effect on her usual weekly hours. Demographic controls are both spouses' ages (wife's entered quadratically), both spouses' college attainment, the number of own children (entered quadratically) and log husband's labor income. Standard errors are clustered on county (430 clusters). *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

TABLE A11. Wife's labor supply by housing tier

Tier	Couples	LFP	Weekly hrs
Renter	715,502	68.66%	25.83
Owner, mortgaged	1,604,202	76.07%	29.42
Owner outright	336,609	64.16%	24.33
Owner outright, top-quartile value	99,065	60.07%	22.99

Notes: Household-weighted means, unconditional and without covariates. The sixteen-point spread between mortgaged owners and debt-free owners of high-value homes is the same leverage/wealth distinction that appears in the regression estimates of Table A10. Outright owners are older by construction.